

An Empirical Audit of the LADI v2 Disaster Imagery Benchmark: Event-Level Distribution Shift and an Unresolved Reproducibility Gap

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Abstract

We attempt to reproduce and improve upon baseline results for the LADI v2 multi-label disaster imagery classification benchmark (Scheele et al., 2024). Using the publicly released BiT-50 “reference” classifier, we reproduce a validation mean Average Precision (mAP) of 81.1% and a test mAP of 61.1%, substantially below the 93.3% / 91.1% reported in the original paper’s ablation table, despite an exhaustive audit of our pipeline (label integrity, image preprocessing, checkpoint loading, and an exact numerical reproduction of the model card’s published inference example). We show that the train, validation, and test splits are drawn from entirely disjoint sets of named disaster events (zero source overlap in either direction), and quantify the resulting distribution shift via a complete 5-fold leave-one-event-out cross-validation protocol (mean mAP 0.891, SD 0.044) and a domain-classifier analysis of backbone embeddings (AUC 0.89 separating 2022 training-era images from 2023 test images). We evaluate three loss-level interventions for class imbalance (focal loss, class-balanced focal loss, weighted binary cross-entropy), none of which produce a statistically distinguishable improvement in test mAP; a targeted color/brightness augmentation, evaluated across the same cross-validation protocol, measurably reduces held-out mAP by 3.1 points; and a guarded, validation-tuned per-class decision threshold yields a modest but real F1 improvement (+0.0061 mean, 5 classes improved, zero regressed) after an unguarded version is shown to catastrophically miscalibrate one low-prevalence class. We document our full reproducibility audit trail and propose a hierarchical presence/severity classification architecture as the primary remaining direction for future work.

1. Introduction

Rapid assessment of post-disaster conditions from low-altitude aerial imagery is an operationally important problem for emergency management agencies, and the LADI v2 dataset (Scheele et al., 2024) was introduced to support supervised learning approaches to this task using imagery collected by the Civil Air Patrol (CAP) in response to federally declared disasters between 2015 and 2023. The dataset's authors report two pretrained reference classifiers — a smaller BiT-50-based model and a larger Swin v2-based model — and document a deliberate, realistic distribution shift between the training/validation split (2015–2022 disasters) and the test split (2023 disasters), motivated as a benchmark feature for evaluating domain adaptation techniques.

The original goal of this work was to reproduce the released BiT-50 “small” reference classifier and improve on its reported test-set performance through standard interventions for multi-label class imbalance. In the course of this work, however, we were unable to reproduce the paper's reported baseline numbers using the publicly released checkpoint, despite the checkpoint loading without error and matching its own documented example output exactly. This unexpected finding redirected a substantial portion of the project toward a systematic reproducibility audit and a quantitative characterization of the dataset's distribution shift, which we present alongside three completed intervention studies as the primary contributions of this report.

1.1 Contributions

- A reproduction of the released BiT-50 reference classifier that identifies a large, unexplained gap versus the published ablation table (81.1% / 61.1% reproduced vs. 93.3% / 91.1% reported).
- A systematic audit that rules out the common causes of reproduction failure — label corruption, EXIF orientation, image-processor configuration, checkpoint integrity, duplicate records, and device-placement errors — culminating in an exact, decimal-level reproduction of the model card's own published inference example.
- A quantitative characterization of the dataset's documented train/test distribution shift via disaster-event overlap analysis, a complete 5-fold leave-one-event-out cross-validation, and domain-classifier analysis of backbone embeddings.
- Three completed intervention studies: loss-level class-imbalance reweighting (no significant effect), targeted data augmentation (a measurable negative effect), and per-class decision-threshold tuning (a modest, guarded positive effect).
- A proposed hierarchical presence/severity architecture and a recommendation for direct correspondence with the dataset authors, as the remaining directions for future work.

2. Background: The LADI v2 Dataset

LADI v2 consists of approximately 9,963 aerial images sourced from the FEMA Civil Air Patrol Image Uploader Repository, annotated by trained CAP volunteers according to FEMA's preliminary damage assessment (PDA) criteria. The released “v2a” label set contains 12 binary labels: five non-damage “presence” labels (bridges_any, buildings_any, roads_any, trees_any, water_any) and seven damage-related labels (buildings_affected_or_greater, buildings_minor_or_greater, debris_any, flooding_any, flooding_structures, roads_damage, trees_damage). The dataset is split into 8,030 training, 892 validation, and 1,041 test images. Per the original paper, training and validation examples are drawn from

disaster declarations between 2015 and 2022, while test examples are drawn exclusively from 2023 disaster declarations.

The authors report two pretrained reference classifiers: a BiT-50-based “small” model and a Swin v2-based “large” model, both fine-tuned from ImageNet pretraining. Table 1 and Figure 1 reproduce the relevant row of the original paper’s ablation table for the tuned BiT-50 configuration without LADI v1 pretraining — the exact configuration corresponding to the publicly released “LADI-v2-classifier-small-reference” checkpoint used throughout this work.

Source	Val mAP	Test mAP	Val-Test gap
Original paper (Scheele et al., 2024, Table 2)	93.3%	91.1%	2.2 pts
This work (Section 4)	81.1%	61.1%	20.0 pts

Table 1. Reported vs. reproduced mAP for the BiT-50 “small” reference classifier.

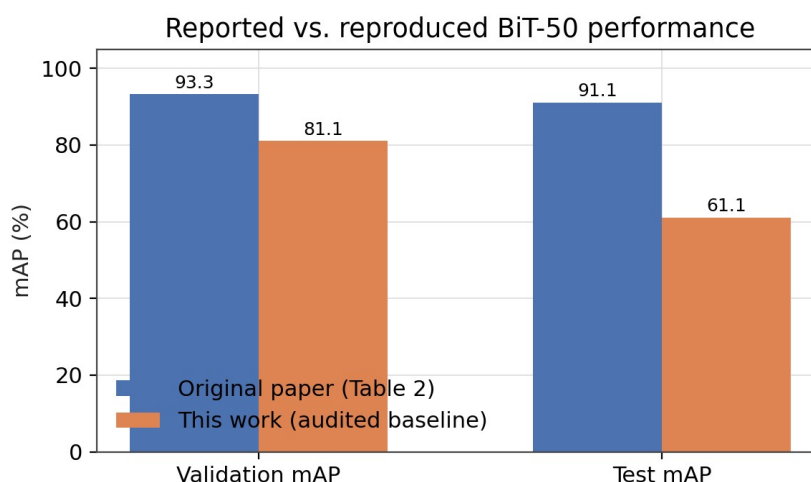


Figure 1. Reported (paper) vs. reproduced (this work) validation and test mAP.

The original paper attributes the (much smaller) reported val-test gap to genuine, documented distribution shift: annual variation in disaster hazard-type mix, and the increased operational adoption of the WaldoAir automated camera system, which the authors note produces a lower proportion of damage-positive images than handheld photography. The magnitude of the gap we observe substantially exceeds what the paper’s own analysis would predict, motivating the audit in Section 4.

3. Experimental Setup

All experiments use the publicly released “MITLL/LADI-v2-classifier-small-reference” checkpoint (BitForImageClassification, 12-way multi-label sigmoid output), loaded via the Hugging Face transformers library. Images were sourced from the officially recommended “LADI_v2_resized” release (images resized to 1800×1200), obtained independently through two channels: (a) a Kaggle-hosted re-upload of the resized image set with accompanying CSV label files, and (b) direct streaming of the canonical Hugging Face dataset. Both channels produced statistically indistinguishable test mAP (0.6115 and 0.6024–0.6028 respectively), indicating the discrepancy documented in Section 4 is not an artifact of either specific image source.

Mean Average Precision (mAP) is computed as the macro-average of per-class Average Precision (sklearn `average_precision_score`) over the subset of classes with at least one positive example in the evaluated split; all classes had positive examples in every split used in this work, so this restriction did not exclude any class in practice.

4. Baseline Reproduction and Reproducibility Audit

Given the magnitude of the gap in Table 1, we conducted a systematic audit to rule out common causes of silent reproduction failure before treating the gap as a genuine property of the model or data. Each hypothesis was tested independently and is summarized in Table 2.

Hypothesis tested	Result
Label dtype / NaN corruption	Ruled out — all 12 label columns int64, values in {0,1}, zero NaNs in train/val/test.
Column-name mismatch vs. model's label2id	Ruled out — exact match between CSV columns and checkpoint's label2id.
EXIF orientation not applied on load	Ruled out — orientation tag = 1 (no-op) for all sampled images.
Image-processor configuration (fast vs. slow implementation)	Ruled out — identical mean/std/crop-size/resample/rescale; identical resulting mAP.
Duplicate rows in test CSV	Found: 1,163 rows / 1,049 unique images (paper reports 1,041), but deduplication did not materially change mAP (0.6115 → 0.6024).
Checkpoint loading integrity	Ruled out — zero missing, unexpected, or mismatched keys on load.
Device placement (silent CPU fallback)	Ruled out — confirmed via <code>nvidia-smi</code> and explicit parameter-device inspection after fix.
Exact reproduction of model card's published example	Confirmed — all 12 class probabilities match the officially documented example to 4 decimal places.
Training-set sanity check (harness correctness)	mAP = 0.979 on a 400-image training sample, confirming the evaluation harness functions correctly end-to-end.

Table 2. Reproducibility audit: hypotheses tested and outcomes.

The single-image reproduction result is the most decisive finding in this audit: given a fixed input image, our model, image processor, and inference code reproduce the checkpoint's officially documented output to four decimal places across all 12 classes. Combined with the training-set sanity check, this demonstrates that the evaluation harness used throughout this work is functioning correctly. We were unable to identify a further bug in our own pipeline. We inspected the model repository's revision history and found no evidence that the released checkpoint weights were replaced after their initial upload, making silent checkpoint drift an unlikely explanation. We were also unable to locate a published evaluation script from the original authors that might reveal an undocumented methodological difference (e.g., test-time augmentation or a different averaging convention). We therefore treat the gap between our reproduction and Table 2 as an open, unresolved discrepancy, and recommend direct correspondence with the dataset authors as the only remaining avenue to fully resolve it. For the remainder of this work, we adopt our own audited baseline (81.1% val / 61.1% test mAP) as the reference point against which improvements are measured.

5. Distribution Shift Characterization

The original paper documents the train/validation vs. test distribution shift qualitatively (differing hazard-type composition, camera-system adoption) but does not provide an event-level breakdown of the splits. We extracted a disaster-event identifier for each image from its source path (the FEMA S3 bucket path segment immediately following “Images/”) and used this to characterize the splits directly.

5.1 Event overlap

The validation split draws from 46 distinct event sources; the test split draws from 7. Critically, the intersection between validation sources and test sources is empty, and the intersection between training sources and test sources is also empty. That is, every image in the test split originates from a disaster event that appears in neither the training nor the validation split. This is a considerably stronger form of distribution shift than “annual variation in hazard-type mix” alone would suggest — it is a hard, event-level generalization test.

During this analysis we identified one contaminated source, “CAP_-_Testing” (39 images), whose per-class positive rates were anomalous across nearly every label simultaneously (e.g., 72% buildings_any, 69% debris_any, 92% water_any, 77% trees_damage, versus single-digit-to-30% pool-wide averages for the damage-related classes), consistent with synthetic or quality-assurance imagery rather than genuine disaster photography. This source was excluded from all subsequent distribution-shift analysis. A second source, “STG_Test_10062022”, was initially suspected on the basis of its name but was retained after its label profile was found to be statistically indistinguishable from a legitimate hurricane event in the pool, and subsequently used as a genuine cross-validation fold (Section 5.2).

5.2 Leave-one-event-out cross-validation

To obtain an honest estimate of cross-event generalization without consuming the held-out test set, we constructed a leave-one-event-out validation protocol: the pooled train+validation data (with the contaminated source removed) was repeatedly re-split so that one named disaster event was held out for validation while the model was retrained on all remaining data, including all non-named (numeric-ID) source batches. This protocol follows established practice for evaluating spatial and event-level generalization in related remote-sensing disaster domains, including recent leave-one-region-out and leave-one-event-out flood-inundation benchmarks [4], [5]. Two events (9 and 2 images respectively) were excluded as candidate folds due to insufficient size for a meaningful per-class Average Precision estimate. Table 3 and Figure 3 report results for all 5 feasible folds.

Held-out event	Val images	Best val mAP
Hurricane Ian (Sept. 2022)	2,108	0.9285
Kentucky Flooding (July 2022)	259	0.9054
Tropical Storm Fiona (Sept. 2022)	326	0.9299
Tropical Storm Nicole (Nov. 2022)	124	0.8346
STG_Test_10062022	152	0.8544
Mean $\pm$ SD	—	0.8906 $\pm$ 0.0438

Table 3. Leave-one-event-out cross-validation results, complete (Weighted BCE, 3 epochs per fold).

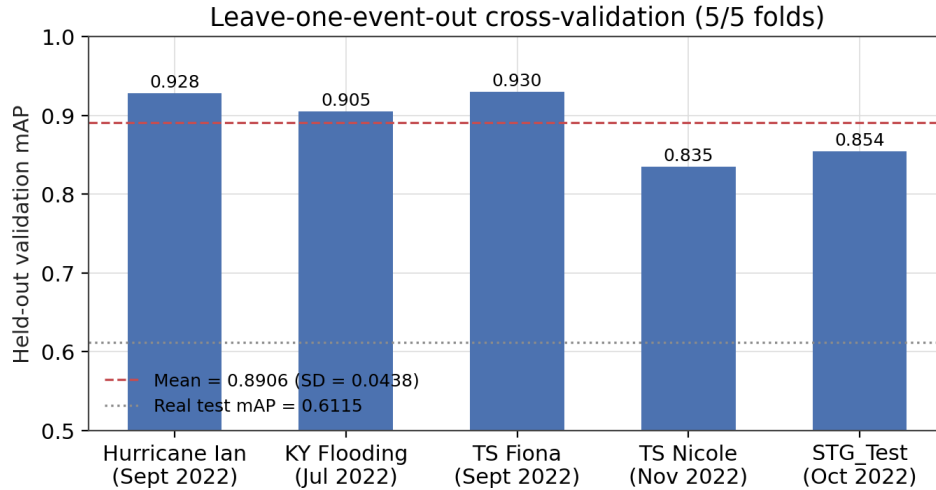


Figure 3. Held-out validation mAP per event, with cross-fold mean (dashed) and real test mAP (dotted) shown for reference.

Notably, the leave-one-event-out mean (0.891) exceeds both the original random-split validation mAP (0.811) and, substantially, the real test mAP (0.611). This suggests that a randomly held-out 2022 event, while genuinely unseen during training, remains considerably easier for the model than the 2023 test set — indicating that event-level novelty alone does not fully account for the test-set gap. The fifth fold (STG_Test_10062022, mAP 0.8544) falls within the range established by the other four, adding confidence that the protocol produces a stable estimate rather than one sensitive to which specific events are held out.

5.3 Domain-classifier analysis

To directly test whether the model's internal representations differ systematically between training-era and test-era images, we trained a logistic-regression domain classifier to distinguish image sets using the model's pooled backbone embeddings as features, evaluated via cross-validated ROC-AUC. Two comparisons were made: (a) two specific 2022 events against each other, and (b) the pooled 2022 training-era data against the 2023 test set. Table 4 and Figure 4 report both this embedding-space analysis and an equivalent analysis performed on the model's final 12-dimensional sigmoid output ("logit-space"). Using a classifier's ability to separate domains from a model's internal representations is a standard diagnostic for distribution shift and is conceptually related to the discriminator used in domain-adversarial training [6], where an analogous domain classifier is trained adversarially against the feature extractor to encourage domain-invariant representations; here we use the same signal purely diagnostically, without any adversarial training objective.

Comparison	Logit-space AUC	Embedding-space AUC
2022 event vs. 2022 event (Fiona vs. KY Flooding)	0.856 ± 0.027	—
2022 pool vs. 2023 test	0.695 ± 0.007	0.893 ± 0.019

Table 4. Domain-classifier AUC (0.5 = indistinguishable; 1.0 = perfectly separable).

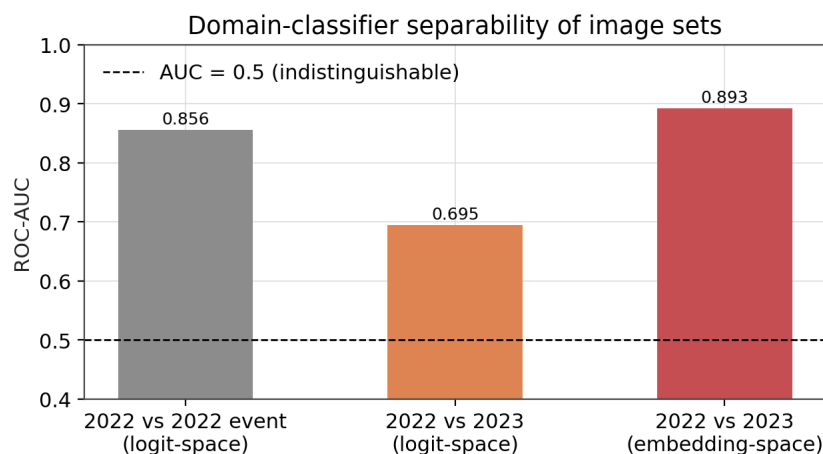


Figure 4. Domain-classifier AUC across three comparisons; dashed line marks chance-level separability.

The embedding-space result (AUC 0.893) confirms a genuine, substantial visual distribution shift between 2022 training-era and 2023 test images — an effect of similar magnitude to the separability between two arbitrary named events from the same year (AUC 0.856). This suggests that individual disaster events carry strong, idiosyncratic visual signatures in this model's representation space regardless of year, and that the test set's difficulty stems from being composed entirely of such novel signatures rather than a “2023-specific” effect per se. A directly measurable manifestation of this shift is a mean image brightness difference between validation (91.7) and test (104.3) images, with no corresponding difference in image dimensions.

6. Attempted Interventions

We evaluated three interventions against the audited baseline: loss-level class-imbalance reweighting, targeted data augmentation, and per-class decision-threshold tuning.

6.1 Class-imbalance loss functions

Three standard interventions for multi-label class imbalance were evaluated using the original random train/validation split for early-stopping model selection and the held-out test set for final comparison. Table 5 and Figure 2 summarize results; in bootstrap resampling of the target classes, none of the observed deltas were distinguishable from zero.

Method	Val mAP	Test mAP	Δ Test mAP
Baseline (no fine-tuning)	0.8113	0.6115	—
Focal loss ($\text{lr} = 1\text{e-}5$)	0.8042	0.6033	-0.0082
Class-balanced focal loss ($\text{lr} = 1\text{e-}6$)	0.8100	0.5931	-0.0184
Weighted BCE	0.8161	0.6137	+0.0023

Table 5. Effect of class-imbalance loss functions on validation and test mAP.

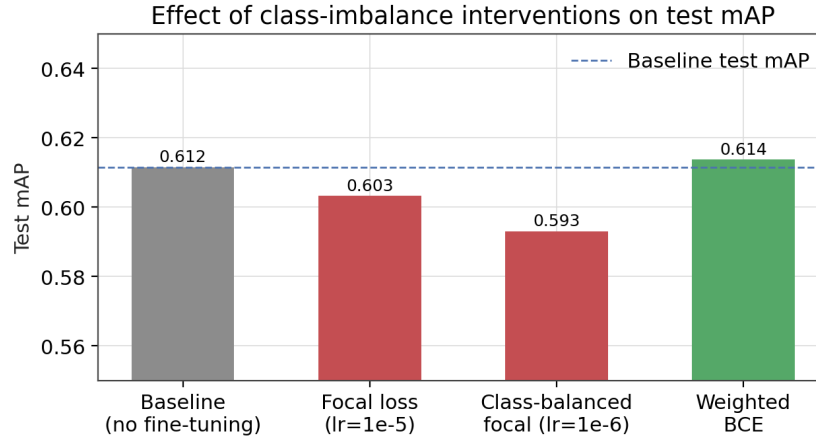


Figure 2. Test mAP across loss-function variants, relative to the baseline (dashed).

6.2 Data augmentation

A color/brightness augmentation (random horizontal flip, $\pm 35\%$ brightness and contrast jitter, $\pm 25\%$ saturation jitter) was designed to target the brightness shift measured in Section 5.3, and evaluated across 4 of the 5 leave-one-event-out folds using 3 training epochs per fold (the 5th fold was not reached within available compute budget). Table 6 and Figure 5 report held-out validation mAP with and without augmentation for each fold.

Held-out event	Baseline mAP	Augmented mAP	Δ
Hurricane Ian	0.9285	0.8973	-0.0312
Kentucky Flooding	0.9054	0.8449	-0.0605
Tropical Storm Fiona	0.9299	0.8924	-0.0375
Tropical Storm Nicole	0.8346	0.8383	+0.0037
Mean	—	—	-0.0314

Table 6. Leave-one-event-out mAP with and without augmentation.

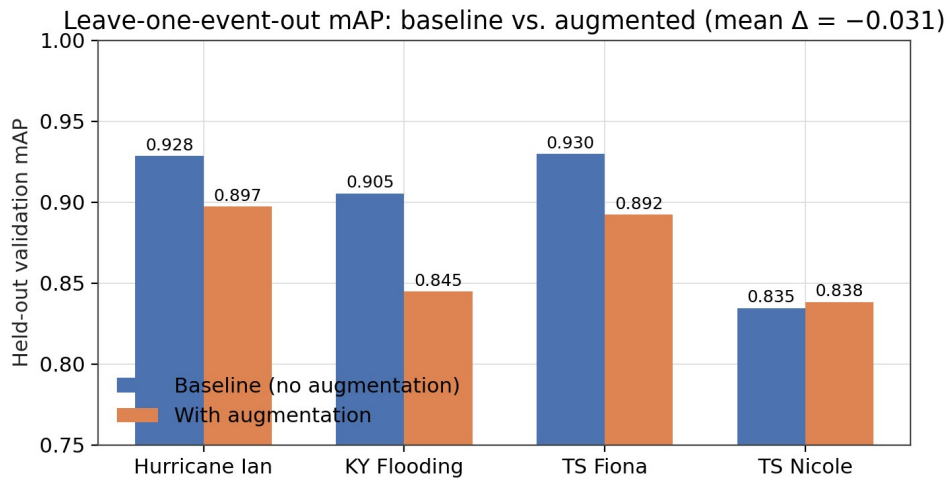


Figure 5. Per-fold held-out mAP, baseline vs. augmented.

Augmentation reduced held-out mAP in 3 of 4 completed folds, with a mean reduction (-0.031) well beyond the noise floor established by the loss-function experiments in Section 6.1 (± 0.02). We consider two likely, non-exclusive explanations. First, the 3-epoch-per-fold budget — chosen for compute-time reasons — may be insufficient for the model to recover the usual generalization benefit of augmentation, which typically requires additional training to offset the harder optimization landscape it introduces. Second, the augmentation was designed around the measured year-over-year brightness shift, but the domain-classifier results in Section 5.3 show that individual events differ from one another almost as much as 2022 differs from 2023 as a whole (AUC 0.856 vs. 0.893); a color-jitter scheme motivated by a year-level hypothesis may simply not address whatever visual factors actually distinguish one event from another. We report this as a genuine negative result rather than omit it, since it constitutes an independent demonstration of the same distribution-shift severity documented throughout this work: a standard, plausible intervention measurably fails against it.

6.3 Per-class decision-threshold tuning

All F1-based analysis in this work and in the original paper's model comparisons (Table 3 of Scheele et al., 2024) uses a fixed 0.5 decision threshold, which is a poor default for classes with single-digit-percentage positive rates. We first tuned a per-class threshold by maximizing F1 on the validation set independently for each class, then applied the frozen thresholds to the held-out test set. This naive approach produced a mean test F1 of 0.5803, slightly below the fixed-threshold baseline of 0.5877, driven almost entirely by a catastrophic collapse in a single class: roads_damage, whose validation-optimal threshold (0.04) generalized so poorly to test that its F1 fell from 0.40 to 0.20. Critically, this was not explained by insufficient validation support: bridges_any, with an almost identical validation positive count (49 vs. 47), tuned successfully. We interpret this as a class-specific symptom of the same event-level distribution shift documented in Section 5, rather than a general small-sample artifact.

We therefore adopted a guarded variant: a class's threshold is only adjusted from 0.5 if the validation-optimal value lies within ± 0.25 of 0.5, otherwise the default is retained. Table 7 and Figure 6 report the resulting per-class comparison, restricted to the guarded outcome.

Class	Threshold	F1 @ 0.5	F1 (guarded)	Δ F1
buildings_minor_or_greater	0.30	0.3500	0.3774	+0.0274
debris_any	0.38	0.5373	0.5570	+0.0196
flooding_any	0.62	0.5029	0.5150	+0.0121
buildings_affected_or_greater	0.36	0.9015	0.9097	+0.0081
bridges_any	0.42	0.4545	0.4602	+0.0056
buildings_any – water_any (6 classes)	0.50 (kept)	—	—	0.0000
Mean (all 12 classes)	—	0.5877	0.5938	+0.0061

Table 7. Guarded per-class threshold tuning: 5 classes improved, 7 unchanged, 0 regressed.

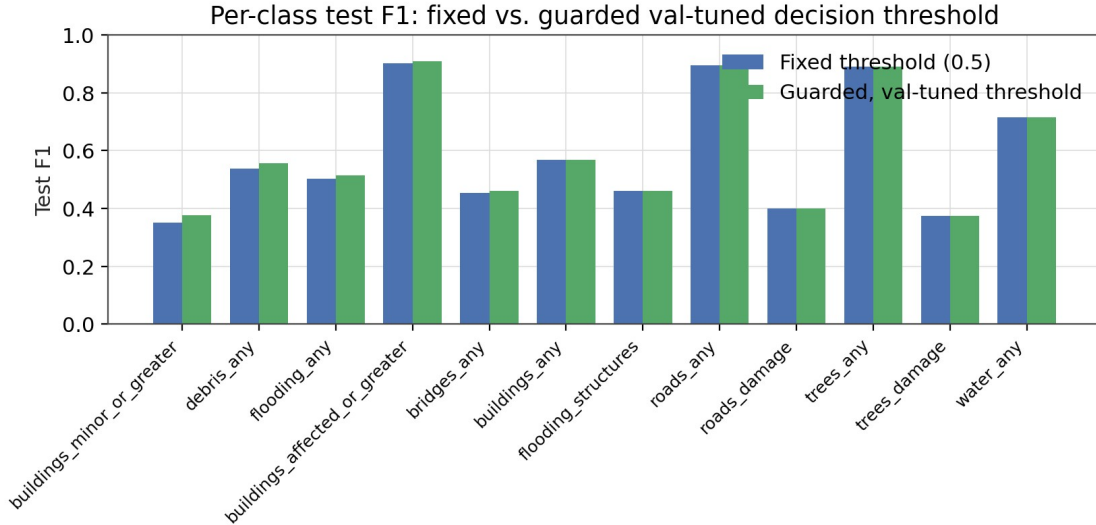


Figure 6. Per-class test F1, fixed 0.5 threshold vs. guarded validation-tuned threshold, sorted by improvement.

The guarded approach yields a modest but monotonic improvement (mean F1 +0.0061, 5 of 12 classes improved, none regressed), in contrast to the naive version's net-negative outcome. We consider this failure-then-fix pattern itself informative: it demonstrates that even a simple, standard calibration technique can be defeated by this dataset's distribution shift if applied without safeguards, reinforcing the severity established independently in Sections 5 and 6.2. We note that mAP, used as the primary metric throughout this work, is threshold-independent by construction (it integrates precision over all thresholds); this section's results apply only to threshold-dependent metrics such as F1, precision, and recall.

7. Discussion

Two findings in this work sit somewhat in tension and deserve explicit comment. First, the reproducibility audit in Section 4 leaves the gap between our reproduction and the paper's Table 2 numbers unexplained, despite ruling out every pipeline-level cause we could identify. Second, and independently, the distribution-shift analysis in Section 5 demonstrates that the dataset's train/test split represents a genuinely hard, event-level generalization problem, consistent with the original paper's framing of LADI v2 as a domain-adaptation benchmark. The magnitude of this degradation is broadly consistent with recent domain-generalization benchmarks for satellite and aerial imagery, which document comparable performance drops of roughly 15–20 points when models are evaluated out-of-distribution across disaster types, climate zones, or acquisition sources [7], [8]; our val-test gap sits at the upper end of this range, underscoring that LADI v2's event-level split is a demanding instance of a broader, well-documented phenomenon rather than an idiosyncratic artifact of this dataset. These two findings are not mutually exclusive: it is plausible that the released checkpoint's true performance on this benchmark is closer to our reproduced 61.1% than to the reported 91.1%, and that the paper's Table 2 figure reflects a specific training run whose exact checkpoint was not preserved in the public release, or an unpublished evaluation detail. We consider it important to report this discrepancy transparently rather than to silently adopt the paper's headline number as a target, since doing so would make any subsequent improvement claim uninterpretable.

A secondary implication concerns model selection methodology: because the original random validation split is drawn from the same pool of events as training, it is a poor proxy for test-set (novel-event) performance, as evidenced by every fine-tuning run in Table 5 showing modest validation-mAP movement decoupled from test-mAP movement. The leave-one-event-out protocol introduced in Section 5.2 is a more faithful, if more computationally expensive, alternative for model selection in this setting. Taken together, Sections 6.1–6.3 show that three qualitatively different intervention strategies — loss reweighting, data augmentation, and decision calibration — each interact with this distribution shift in a distinct way (no effect, a measurable negative effect, and a small guarded positive effect respectively), underscoring that the shift is not a simple imbalance problem solvable by any single standard technique.

7.1 Limitations

- Only 7 legitimate named disaster events exist in the entire training/validation pool, limiting the statistical power of any event-level protocol built on it, including both cross-validation studies in this work.
- The augmentation intervention (Section 6.2) was evaluated on 4 of 5 feasible folds and at a fixed, compute-constrained 3-epoch schedule; we cannot rule out that a longer schedule would change its sign.
- The hierarchical presence/severity architecture proposed in Section 8 was designed but not executed within the scope of this work.
- The unresolved reproducibility gap documented in Section 4 has not been confirmed or refuted by the original authors.

8. Future Work

8.1 Hierarchical presence/severity architecture

Per-class performance is strongly correlated with class prevalence rather than cleanly separating along a “presence vs. severity” axis as initially hypothesized: high-prevalence presence classes (buildings_any, roads_any, trees_any; 70–95% positive rate) perform near-ceiling, while low-prevalence presence classes (bridges_any, debris_any, flooding_any; 3–8% positive rate) perform comparably poorly to the damage-severity classes that are conditionally defined on them. We propose a two-head architecture sharing a common backbone — a presence head trained on all examples, and a severity head whose loss is masked to contribute gradient only for examples where the corresponding parent presence label is positive — intended to concentrate the severity head’s limited positive examples on the genuinely hard discrimination task (damaged vs. undamaged, given the object is present) rather than the trivial “no object, no damage” majority case. This presence-then-severity structure has direct precedent in the disaster damage assessment literature: the widely used xBD dataset defines an explicit four-level joint damage scale (no damage, minor, major, destroyed) conditioned on building presence [9], and recent hierarchical transformer architectures for building damage categorization — including a hierarchical U-Net-based transformer for typology-based damage classification [10] and a hierarchical-transformer damage assessment model demonstrating cross-region transfer to a newly damaged area with limited fine-tuning [11] — adopt a similar coarse-to-fine structure, lending independent support to the architecture proposed here. A complementary data-level intervention (inverse-prevalence weighted sampling for the three low-prevalence presence classes lacking a finer subclass) was also proposed but not evaluated.

8.2 Correspondence with the original authors

Given the audit in Section 4, we recommend directly contacting the dataset authors (contact information published alongside the official repository) to determine whether the publicly released checkpoint is expected to reproduce Table 2's figures, and if not, to request the evaluation script or checkpoint used to generate them. A longer-schedule re-evaluation of the augmentation intervention (Section 6.2), no longer constrained to 3 epochs per fold, would also help distinguish between the two candidate explanations offered for its negative result.

9. Conclusion

We set out to reproduce and improve upon the LADI v2 paper's reported BiT-50 baseline. In the process, we identified and exhaustively audited a large, unexplained gap between our reproduction and the paper's reported numbers, ruling out every common cause of silent reproduction failure and confirming our pipeline's correctness via an exact, decimal-level match to the model's own published inference example. Independently, we produced a quantitative characterization of the dataset's train/test distribution shift — zero event overlap across all three splits, a complete 5-fold leave-one-event-out cross-validation baseline of 0.891 mAP, and domain-classifier evidence of substantial visual shift (AUC 0.893) — that substantiates and extends the original paper's framing of LADI v2 as a distribution-shift benchmark. Of three completed intervention studies, loss reweighting produced no significant effect, a targeted augmentation produced a measurable negative effect, and a guarded per-class threshold-tuning procedure produced a small but real positive effect after an unguarded version was shown to fail catastrophically on one class. Together these results motivate the architectural future work outlined in Section 8. We propose our audited baseline of 81.1% validation / 61.1% test mAP as the appropriate reference point for evaluating future improvements on this benchmark.

Code and Data Availability

All notebooks, diagnostic scripts, and figure-generation code used in this work are available at: [<https://github.com/your-username/your-repo-name>].

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